

the norm of the Jacobian we use a subset of the training data of 2500 examples. Each run requires approximately 5GPU hour on A100.

LLM experiments. We use the NanoGPT repository (Karpathy, 2023) for training GPT-2 models (Radford et al., 2019) on OpenWebText (Gokaslan et al., 2019). All training documents are concatenated in a single stream from which a new batch is sampled with replacement on every iteration of training. We train a 124M parameter model known as GPT-2-small for 50 000 iterations instead of the default 600 000 to make grid searches over the learning rate and weight decay parameters more accessible within an academic budget. We mostly use the context length of 256 for faster experiments except for Fig. 8 where we use the context length of 1024 since we observed that a larger context length is crucial to observe loss divergences with moderate learning rates (such as 0.001 for Adam). We train with AdamW (Loshchilov & Hutter, 2019) using batch size 256, default LR 0.0006 (unless mentioned otherwise), $\beta_1 = 0.9$, $\beta_2 = 0.95$, a short 400-iteration LR warmup, and $10\times$ cosine LR decay. For the runs with SGD with momentum, we use the learning rate 0.3 and momentum parameter 0.9 using the same LR schedule as for AdamW. We initialize all parameters with the standard deviation equal to 0.02. We keep all other hyperparameters at their default values as in the NanoGPT repository. We perform all experiments on A100 Nvidia GPUs that support fast bfloat16 training. Each training run of GPT-2-small for 50 000 iterations takes around 12 hours on a single GPU.

C Weight decay for overparametrized deep networks: additional experiments and details

C.1 A graphical illustration of the fine-tuning phase

Here, we plot an illustrative graphic in Fig. 9 to give an idea of what happens during the fine-tuning phase.

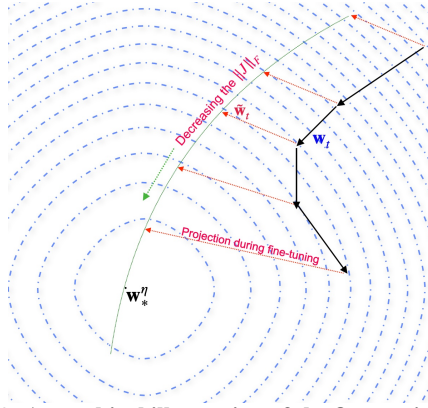


Figure 9: A graphical illustration of the fine-tuning phase.

C.2 Supporting derivations

Here we prove that the scale of noise is well approximated by training loss in the case of binary classification instead of classification in the case of multiple classes. The proof follows the lines of Wojtowysch (2021).

Proposition 2. Assume $\|\mathbf{w}\| \in [a, b]$, for any $x \in \mathcal{D}$, $\|\nabla h(\mathbf{w}, x)\| \in [m, M]$ holds. For n sufficiently large, there exists constants c_1, c_2 such that

$$\mathbb{E} \left[\|g(\mathbf{w})\|^2 \right] \leq c_2 \mathcal{L}(\mathbf{w})$$

Proof. The noise in the case when the gradient is computed at (x_i, y_i) is

$$g(\mathbf{w}) = \ell'(y_i, h(\mathbf{w}, x_i)) \nabla h(\mathbf{w}, x_i) - \frac{1}{n} \sum_i \nabla \ell'(y_i, h(\mathbf{w}, x_i)) \nabla h(\mathbf{w}, x_i),$$

Taking the expectation over uniform sampling over i , we have,

$$\mathbb{E}\|g\|^2 = \frac{1}{n} \sum_{i=1}^n \left(\ell'(y_i, h(\mathbf{w}, x_i)) \right)^2 \|\nabla h(\mathbf{w}, x_i)\|^2 - \frac{1}{n^2} \left\| \sum_i \nabla \ell'(y_i, h(\mathbf{w}, x_i)) \nabla h(\mathbf{w}, x_i) \right\|^2 \quad (6)$$

Upper bound: Using the self-bounding property of the binary CE, i.e., $(\ell'^2) \leq l$ and $\|\nabla h(\mathbf{w}, x)\|^2 \leq M^2$.

$$\mathbb{E}\|g\|^2 \leq M^2 \frac{1}{n} \sum_{i=1}^n \ell(y_i, h(\mathbf{w}, x_i)) = M^2 \mathcal{L}(\mathbf{w}).$$

□

Comment on the Lower bound: Since the iterates are bound, we can assume there exists a constant c such that $(\ell'^2) \geq cl$. as the second term in 6 is decreasing with $O(n^{-2})$, we can assume that the first term is dominating and relevant and can lower bound the first term as,

$$\mathbb{E}\|g\|^2 \geq cm^2 \frac{1}{n} \sum_{i=1}^n \ell(y_i, h(\mathbf{w}, x_i)) = cm^2 \mathcal{L}(\mathbf{w}).$$

C.3 Additional figures for the over-training regime

In this section, we report additional experimental results related to Section 2 in the main text. In Fig. 10 we report analogous results for the jacobian norm and test error of the EMA for Resnet18 on CIFAR10. In Fig. 11 and 2 we report the train CE for VGG and ResNet18 on CIFAR-10 and ResNet34 trained on CIFAR-100. We can observe how when weight decay is used in combination with large LR, the train CE stabilizes at some approximately constant level.

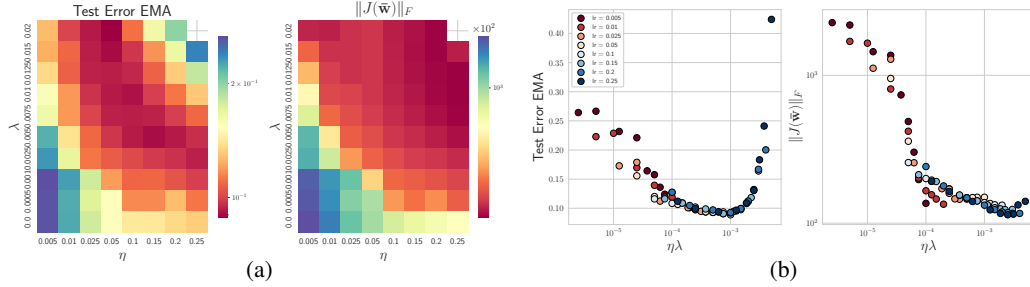


Figure 10: **Resnet18 on CIFAR10** We train Resnet18 on CIFAR10 for 100 epochs with different η and λ . Fig. 10a reports a heatmap of the test error and Jacobian norm for the EMA for all the different combinations of parameters. The test error presents an optimal value of η when λ is fixed and, consistently with conjecture 1, the Jacobian norm decreases monotonically. More over, Fig. 10b shows how the optimality might depend only on the product $\eta\lambda$ for which the test error has a U-shape and the Jacobian norm decreases monotonically.

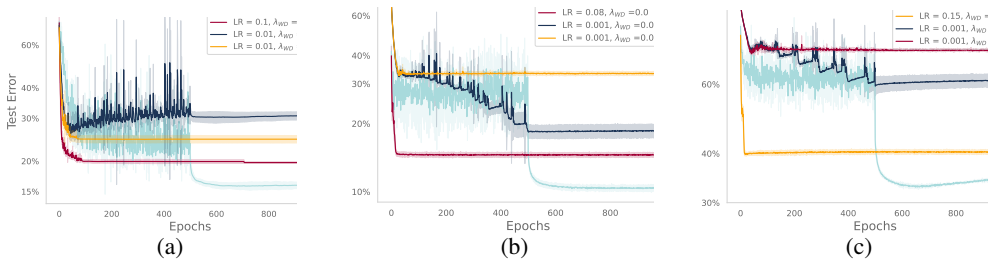


Figure 11: **Training with and w/o weight decay.** We report the test error for VGG (11a) and ResNet (11b, 11c) trained on CIFAR-10/100 with and without weight decay and with small and large learning rates. After the first 500 epochs the learning rate is decayed to $\eta = 10^{-4}$ for all the curves.

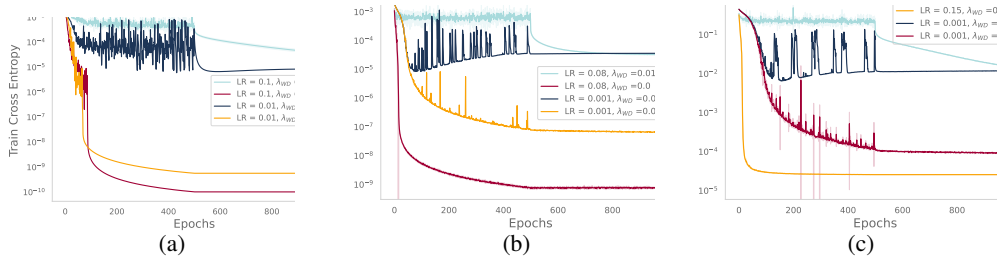


Figure 12: **Training with and w/o weight decay.** We report the train CE for VGG (12a) and ResNet (12b, 12c) trained on CIFAR-10/100 with and without weight decay and with small and large learning rates. After the first 500 epochs the learning rate is decayed to $\eta = 10^{-4}$ for all the curves.

Connection between SGD covariance and Hessian. Much of the literature related to implicit bias relies on the assumption that the covariance of the noise of SGD is strictly related to the hessian of the loss function as discussed in Sec 2. Denoting the Hessian $H(\mathbf{w}) := \nabla^2 \mathcal{L}(\mathbf{w})$ we can write it as the so-called Gauss-Newton decomposition (Sagun et al., 2017; Pappayan, 2018) $H(\mathbf{w}) = G(\mathbf{w}) + E(\mathbf{w})$. To measure the cosine similarity (CS) between $H(\mathbf{w})$ and the covariance Σ_t we compute

$$CS = \mathbb{E} [\cos (H(\mathbf{w})\mathbf{v}, \Sigma_t \mathbf{v})]$$

where $\mathbf{v}$ is sampled from the Gaussian distribution in $\mathbb{R}^P$ and $\cos(\mathbf{u}, \mathbf{v}) = \langle \mathbf{u}, \mathbf{v} \rangle / \|\mathbf{u}\| \|\mathbf{v}\|$. The results are reported in Fig. 13.

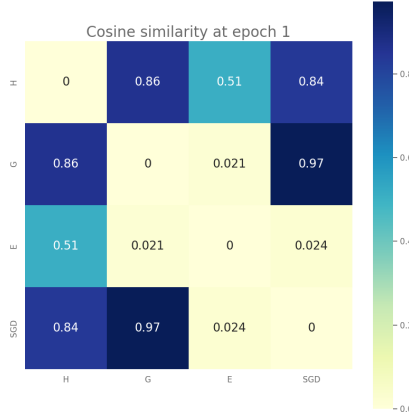


Figure 13: **Cosine similarity between hessian and Noise covariance:** we compute the cosine similarity between the hessian and the covariance of the SGD noise for a scale-invariant ResNet after one epoch with large lr $\eta = 0.005$. The results show how the two matrices are correlated and in particular how the SGD noise covariance is highly correlated with $G(\mathbf{w})$.